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A smaller planar graph without 4-, 5-cycles and intersecting triangles that is not 3-choosable

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ABSTRACT

In 2006, Montassier et al. constructed a planar graph without 4-, 5-cycles and intersecting triangles that is not 3-choosable [M. Montassier, A. Raspaud, W. Wang, Bordeaux 3-color conjecture and 3-choosability, Discrete Math. 306 (2006) 573–579]. It has 506 vertices. In this paper, we construct a smaller planar graph with the same properties. The planar graph has 380 vertices.

1. Introduction

Let G be a graph, $V(G)$ be the vertex set of G, $E(G)$ be the edge set of G and $F(G)$ be the face set of G. If $uv \in E(G)$, we say that u and v are adjacent. A *triangle* in G is synonymous with a 3-cycle. We say that two cycles (or faces) of a graph are adjacent if they share at least one common edge. Two cycles (or faces) are intersecting if they share at least one common vertex.

A *proper k-coloring* of a graph G is a mapping $\sigma : V(G) \to C_k = \{c_1, c_2, \dots, c_k\}$ such that $\sigma(u) \neq \sigma(v)$ for every edge uv of G. G is called *k-colorable* if it admits a proper k-coloring. If each vertex v is assigned a list $L(v)$ of possible colors and G has a proper coloring σ such that $\sigma(v) \in L(v)$ for all $v \in V(G)$, then we say that G is *list L-colorable* or equivalently that σ is a *list L-coloring* of G. If G is list L-colorable for every list assignment with $|L(v)| \geqslant k$ for all $v \in V(G)$, then G is said *k-choosable*.

The concept of list coloring was introduced by Vizing [7] and independently by Erdös et al. [1]. Grötzsch proved that every planar graph without 3-cycles is 3-colorable [3]. In 1976, Steinberg conjectured that every planar graph without 4-cycles and 5-cycles is 3-colorable [6]. In 1995, Voigt gave an example of planar graph without 3-cycles that is not 3-choosable [9], it has 166 vertices. In 2005, Glebov et al. found a graph with the same properties having 97 vertices [2]. In 1993, Voigt gave a planar graph without 4-cycle and 5-cycle that is not 3-choosable [8], and in 2006, Montassier gave a smaller planar graph (with 209 vertices) that has the same properties [4]. In 2006, Montassier et al. gave a planar graph without 4-, 5-cycles and intersecting triangles that is not 3-choosable, it has 506 vertices [5]. In this paper, we construct a smaller planar graph without 4-cycles, 5-cycles and intersecting triangles that is not 3-choosable, it has 380 vertices.

2. Main results

Let $H(s,t)$ be the graph depicted in Fig. 1 and $L_{a,b}$ its list assignment in Fig. 2. It is easy to see that the graph $H(s,t)$ has no 4-cycles, 5-cycles and intersecting triangles.

Theorem 1. *With list assignment $L_{a,b}$, the color-restricted graph $H(s,t)$ has no $L_{a,b}$-coloring.*

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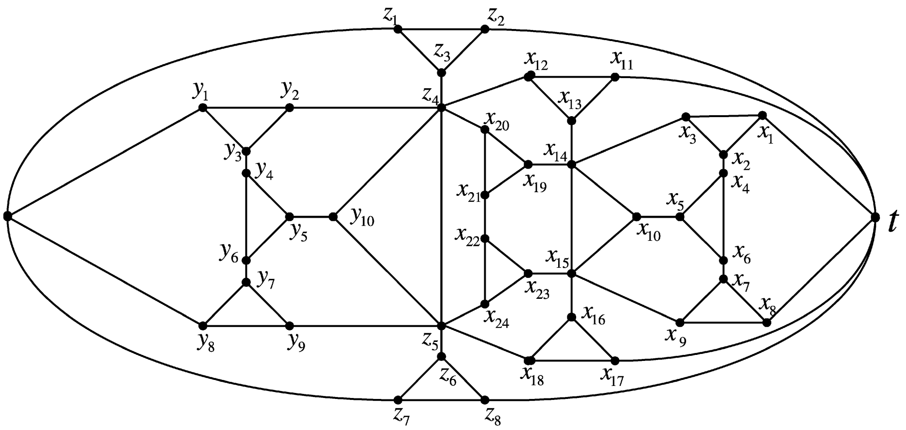

Fig. 1. Graph $H(s,t)$.

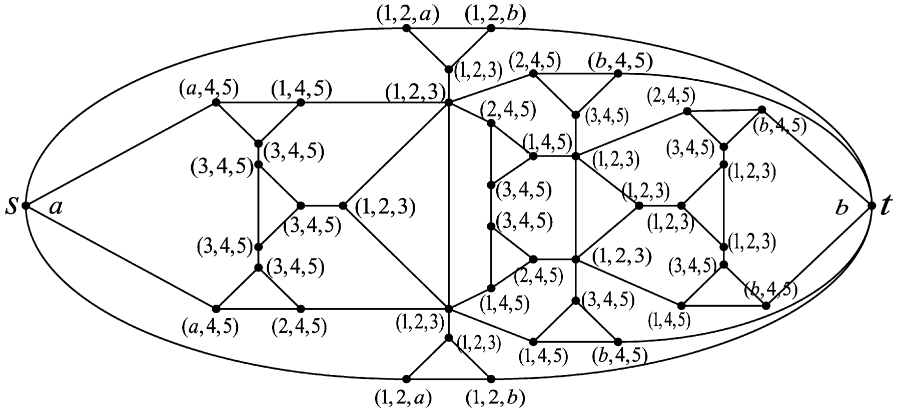

Fig. 2. The list assignment L of the graph $H(s,t)$.

Proof. Let $H = H(s,t)$. Suppose that H has an $L_{a,b}$-coloring σ. We have $\sigma(s) = a$, $\sigma(t) = b$, and hence $\{\sigma(z_1), \sigma(z_2)\} = \{1,2\}$ $(\sigma(z_1) \neq \sigma(z_2))$. Furthermore, $\sigma(z_3) = 3$. By the same reason $\{\sigma(z_7), \sigma(z_8)\} = \{1,2\}$, we have $\sigma(z_6) = 3$, and hence $\{\sigma(z_4), \sigma(z_5)\} = \{1,2\}$ $(\sigma(z_4) \neq \sigma(z_5))$.

Case 1. $\sigma(z_4) = 2$ and $\sigma(z_5) = 1$. Each of vertices x_{11}, x_{12} can be colored only with 4 and 5 ($\sigma(x_{11}) \neq \sigma(x_{12})$), so we have $\sigma(x_{13}) = 3$. By the same reason, each of vertices x_{17} and x_{18} can be colored with 4 and 5 $(\sigma(x_{17}) \neq \sigma(x_{18}))$, so $\sigma(x_{16}) = 3$. Hence, $\{\sigma(x_{14}), \sigma(x_{15})\} = \{1,2\}$ $(\sigma(x_{14}) \neq \sigma(x_{15}))$.

(1) If $\sigma(x_{14}) = 1$ and $\sigma(x_{15}) = 2$, we have $\{\sigma(x_{19}), \sigma(x_{20})\} = \{4,5\}$, and hence $\sigma(x_{21}) = 3$. By the same reason, $\sigma(x_{22}) = 3$. It is a contradiction.
(2) If $\sigma(x_{14}) = 2$ and $\sigma(x_{15}) = 1$, we have $\sigma(x_{10}) = 3$, $\{\sigma(x_3), \sigma(x_1)\} = \{4,5\}$, and hence $\sigma(x_2) = 3$. By the same reason, $\sigma(x_7) = 3$, $\{\sigma(x_4), \sigma(x_6)\} = \{1,2\}$ $(\sigma(x_4) \neq \sigma(x_6))$, and hence $\sigma(x_5)$ cannot be colored (since we already have $\sigma(x_{10}) = 3$).

Case 2. $\sigma(z_4) = 1$ and $\sigma(z_5) = 2$. We have $\sigma(y_{10}) = 3$, $\sigma(y_5) \in \{4,5\}$, $\{\sigma(y_1), \sigma(y_2)\} = \{4,5\}$ $(\sigma(y_1) \neq \sigma(y_2)$, hence $\sigma(y_3) = 3$. By the same reason, $\sigma(y_7) = 3$, $\{\sigma(y_4), \sigma(y_6)\} = \{4,5\}$. So, y_5 cannot be colored (since $\sigma(y_{10}) = 3$).

So planar graph $H(s,t)$ has no $L_{a,b}$-coloring. □

Let $\overline{H}(s',t')$ be the graph (see Fig. 3) obtained from 9 copies of $H(s,t)$, named as $H(s_i,t_j)$ $(1 \leqslant i \leqslant 3,\ 1 \leqslant j \leqslant 3)$ be merging s_i into the new vertex s', merging t_j into the new vertex t'.

Theorem 2. *The graph $\overline{H}(s',t')$ without 4-cycles, 5-cycles and intersecting triangles is not 3-choosable.*

Proof. Let $L(s') = (a_1, a_2, a_3)$ and $L(t') = (b_1, b_2, b_3)$, and let every copy of $H(s_i,t_j)$ in G has list L_{a_i,b_j} shown in Fig. 2 $(1 \leqslant i \leqslant 3,\ 1 \leqslant j \leqslant 3)$. If we color s' and t' with different color $\sigma(s') \in \{a_1, a_2, a_3\}$ and $\sigma(t') \in \{b_1, b_2, b_3\}$, then, in view of Theorem 1, for any coloring of the vertex s' with a_i and t' with b_j $(1 \leqslant i \leqslant 3,\ 1 \leqslant j \leqslant 3)$, there exists a copy $H(s_i,t_j)$ with the list assignment L_{a_i,b_j} which we cannot color properly. Hence $\overline{H}(s',t')$ is not 3-choosable.

It is easy to see that the graph $\overline{H}(s',t')$ has no 4-cycles, 5-cycles and intersecting triangles and it has $42 \times 9 + 2 = 380$ vertices.

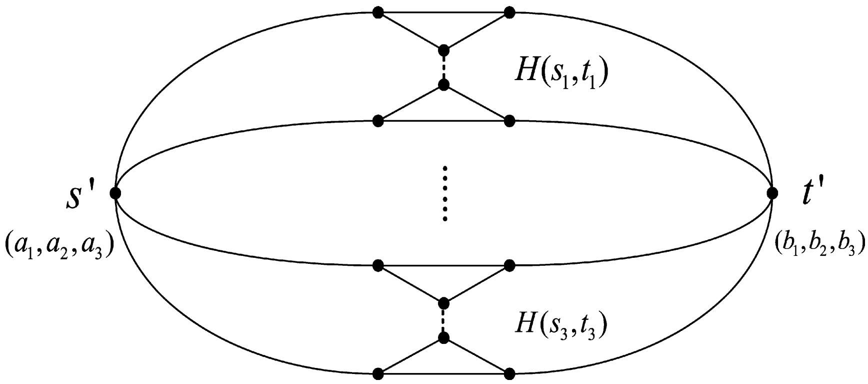

Fig. 3. Graph $\overline{H}(s',t')$.

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